

MID VALLEY, TX, USA

WMO: 722159

Lat: 26.178N

Lon: 97.973W

Elev: 70

StdP: 14.66

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12980

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.7	40.8	16.4	12.7	64.1	22.4	16.9	62.5	26.7	75.7	24.5	75.3	8.5	350	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.5	99.7	75.3	98.4	75.7	96.7	75.9	80.9	87.3	80.2	87.3	79.5	87.2	11.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	153.2	82.2	79.0	151.1	82.2	78.1	146.4	82.2	44.4	86.2	43.6	85.6	43.1	87.0	91.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	20.5	18.7	32.7	103.1	3.2	1.7	30.4	104.3	28.5	105.3	26.7	106.3	24.3	107.6
			29.9	82.2	3.2	3.2	27.6	84.5	25.7	86.4	23.9	88.2	21.6	90.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.6	60.7	65.5	70.9	76.2	81.7	85.6	85.9	87.4	83.2	77.4	69.3	62.5
	DBStd	11.35	9.46	9.88	7.94	5.88	4.61	2.63	2.83	2.47	3.65	6.48	8.45	9.52
	HDD50	52	23	11	2	0	0	0	0	0	0	0	3	13
	HDD65	577	195	108	41	6	0	0	0	0	0	6	55	166
	CDD50	9385	354	445	649	788	982	1068	1111	1161	996	848	581	402
	CDD65	4433	61	121	223	344	517	618	646	696	546	389	182	90
	CDH74	47740	375	785	1546	2997	5501	7635	8109	9058	5904	3850	1417	563
	CDH80	22282	63	215	481	1144	2461	3943	4170	4933	2733	1605	419	115
Wind	WSAvg	8.5	8.1	9.3	9.8	10.0	10.0	9.1	9.4	8.3	6.0	6.8	7.4	7.9
Precipitation	PrecAvg	24.9	0.9	1.0	1.1	1.7	2.3	2.3	2.4	2.0	5.4	3.2	1.4	1.2
	PrecMax	32.0	3.2	3.4	6.4	6.5	6.2	7.2	11.0	7.7	10.7	8.4	5.1	4.9
	PrecMin	15.9	0.1	0.0	0.1	0.0	0.0	0.1	0.0	0.0	1.2	0.0	0.1	0.1
	PrecStd	5.6	0.9	1.0	1.4	1.9	1.7	1.9	3.2	1.8	2.4	2.3	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.9	90.3	92.8	95.4	97.3	100.3	101.4	101.4	99.9	95.0	90.3	86.5
		MCWB	67.6	66.8	71.3	74.5	76.5	74.7	75.1	74.6	74.8	75.0	73.6	70.5
	2%	DB	81.5	85.7	88.1	91.1	94.9	97.9	99.0	99.9	97.0	92.6	87.7	83.2
		MCWB	68.3	69.2	71.1	74.1	76.6	75.7	75.9	75.5	75.3	75.9	72.9	70.0
	5%	DB	78.8	82.3	85.3	89.6	92.9	96.4	96.9	98.7	94.6	90.7	84.7	80.5
		MCWB	66.1	69.1	70.6	72.9	75.9	75.5	75.9	75.9	76.0	74.9	71.1	68.9
	10%	DB	74.8	79.2	82.1	86.8	90.9	94.7	94.7	96.8	91.9	88.5	82.0	77.0
		MCWB	65.2	67.5	69.1	72.0	75.5	75.3	75.5	76.2	76.4	74.0	70.2	68.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.0	73.9	75.4	78.5	81.0	81.9	81.4	81.1	81.6	81.3	77.2	74.5
		MCDB	78.4	82.3	85.9	87.7	89.2	87.0	86.9	87.9	87.9	85.9	85.3	81.4
	2%	WB	71.1	72.6	73.8	77.1	79.3	80.4	80.4	80.3	80.3	79.5	75.4	73.1
		MCDB	77.0	79.9	82.0	86.0	87.5	87.3	86.8	87.7	87.2	85.8	83.0	79.0
	5%	WB	69.5	71.4	72.9	75.9	78.4	79.6	79.6	79.7	79.3	78.3	74.1	71.6
		MCDB	75.3	77.9	80.2	84.1	86.6	87.5	86.8	87.6	86.7	85.3	80.5	77.5
	10%	WB	67.7	70.2	71.8	74.7	77.6	78.8	79.0	79.0	78.4	77.2	72.8	69.9
		MCDB	73.2	76.0	78.3	82.3	85.8	87.1	86.6	87.4	86.1	84.5	78.9	75.2
Mean Daily Temperature Range	5% DB	MDBR	19.6	19.6	18.9	19.5	17.8	18.6	18.1	19.5	17.9	20.2	20.0	18.9
		MCDBR	21.7	21.2	20.1	20.8	18.8	20.3	20.2	21.1	21.1	20.2	20.4	21.0
		MCWBR	10.7	9.1	7.3	7.7	5.2	4.4	4.3	4.3	5.2	6.5	7.8	10.6
	5% WB	MCDBR	18.7	17.9	17.0	18.0	16.6	17.2	16.7	17.8	16.4	16.9	18.1	18.0
		MCWBR	11.2	9.0	7.4	7.1	5.6	4.9	4.8	4.6	5.2	7.2	8.9	11.0
Clear-Sky Solar Irradiance	taub		0.352	0.369	0.394	0.425	0.431	0.416	0.431	0.421	0.427	0.382	0.370	0.354
	taud		2.479	2.422	2.352	2.279	2.298	2.395	2.359	2.396	2.346	2.466	2.451	2.479
	Ebn at Noon		285	288	286	278	275	277	272	276	272	280	276	279
	Edh at Noon		30	35	39	43	43	38	40	38	39	33	31	30
All-Sky Solar Radiation	RadAvg		1000	1212	1483	1751	1912	2115	2102	2014	1670	1456	1125	905
	RadStd		110	181	167	155	129	78	146	90	125	140	121	127

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189

Nomenclature: See separate page